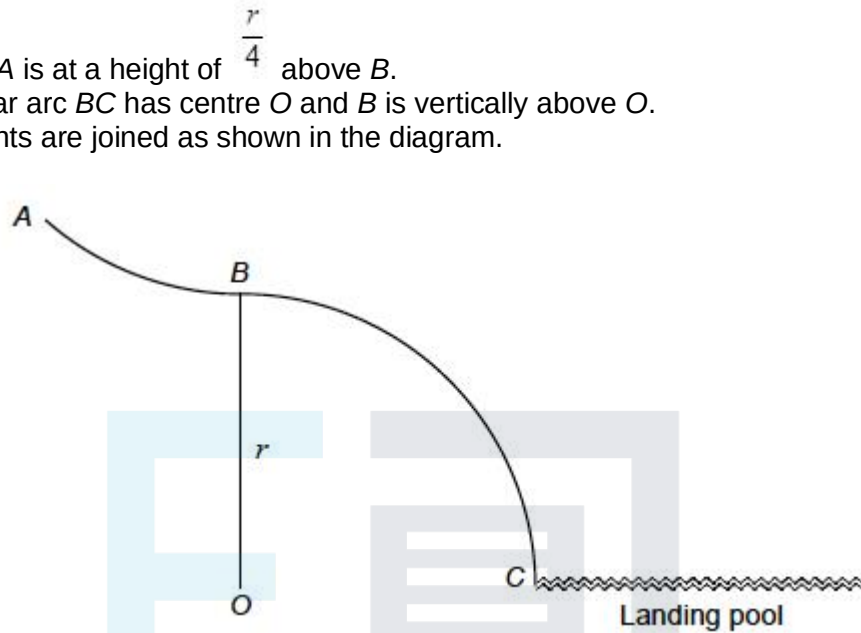


Q1.

The diagram shows part of a water park slide, ABC .

The slide is in the shape of two circular arcs, AB and BC , each of radius r .

The point A is at a height of $\frac{r}{4}$ above B .
 The circular arc BC has centre O and B is vertically above O .
 These points are joined as shown in the diagram.



A child starts from rest at A , moves along the slide past the point B and then loses contact with the slide at a point D .

The angle between the vertical, OB , and OD is θ

Assume that the slide is smooth.

- (a) Show that the speed v of the child at D is given by $v = \sqrt{\frac{gr}{2}(5 - 4\cos\theta)}$, where g is the acceleration due to gravity. (3)
- (b) Find θ , giving your answer to the nearest degree. (2)
- (c) A refined model takes into account air resistance. Explain how taking air resistance into account would affect your answer to part (b). (2)
- (d) In reality the slide is not smooth. It has a surface with the same coefficient of friction between the slide and the child for its entire length.

Explain why the frictional force experienced by the child is not constant.

(1)

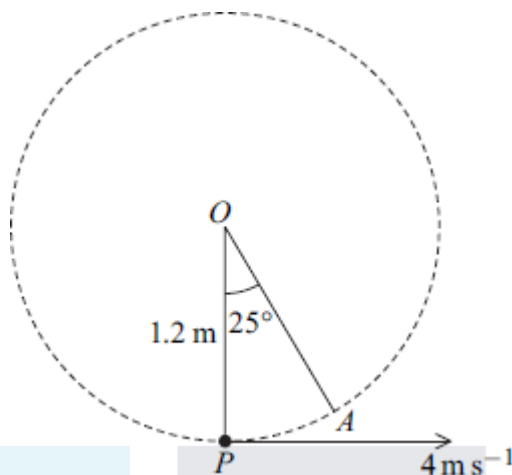
(Total 8 marks)

Q2.

A small ball, of mass 3 kg, is suspended from a fixed point O by a light inextensible string

of length 1.2 m. Initially, the string is taut and the ball is at the point P , vertically below O . The ball is then set into motion with an initial horizontal velocity of 4 m s^{-1} .

The ball moves in a vertical circle, centre O . The point A , on the circle, is such that angle AOP is 25° , as shown in the diagram.

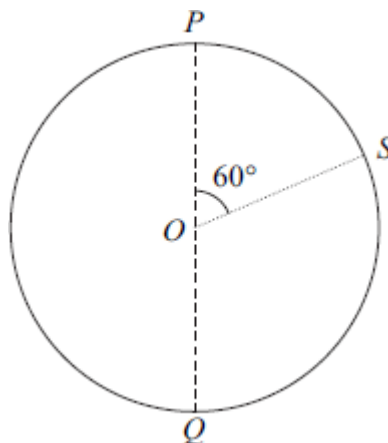


- (a) Find the speed of the ball at the point A . (4)
- (b) Find the tension in the string when the ball is at the point A . (3)
- (Total 7 marks)

Q3.

A bead, of mass m , moves on a smooth circular ring, of radius a and centre O , which is fixed in a vertical plane. At P , the highest point on the ring, the speed of the bead is $2u$; at Q , the lowest point on the ring, the speed of the bead is $5u$.

- (a) Show that $u = \sqrt{\frac{4ag}{21}}$. (4)
- (b) S is a point on the ring so that angle POS is 60° , as shown in the diagram.



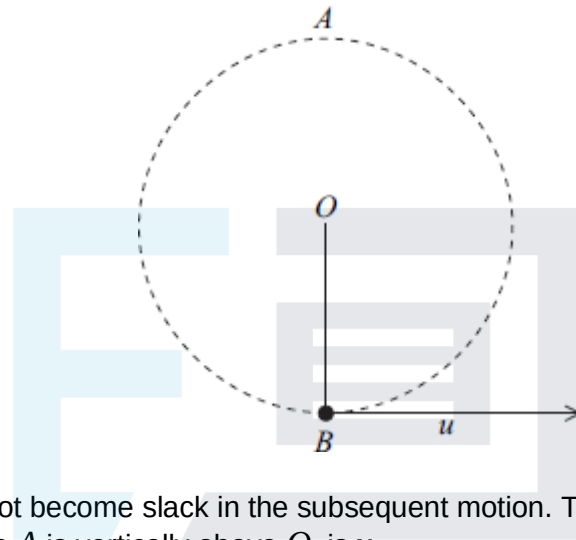
Find, in terms of m and g , the magnitude of the reaction of the ring on the bead when the bead is at S .

(5)

(Total 9 marks)

Q4.

A small bead, of mass m , is suspended from a fixed point O by a light inextensible string of length a . With the string taut, the bead is at the point B , vertically below O , when it is set into vertical circular motion with an initial horizontal velocity u , as shown in the diagram.



The string does not become slack in the subsequent motion. The velocity of the bead at the point A , where A is vertically above O , is v .

- (a) Show that $v^2 = u^2 - 4ag$.

(2)

- (b) The ratio of the tensions in the string when the bead is at the two points A and B is $2 : 5$.

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- (i) Find u in terms of g and a .

(7)

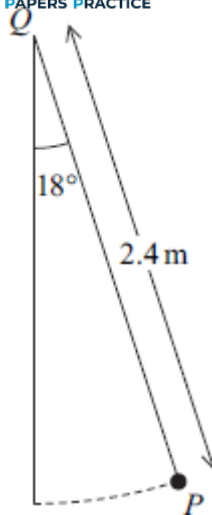
- (ii) Find the ratio $u : v$.

(2)

(Total 11 marks)

Q5.

Simon, a small child of mass 22 kg , is on a swing. He is swinging freely through an angle of 18° on both sides of the vertical. Model Simon as a particle, P , of mass 22 kg , attached to a fixed point, Q , by a light inextensible rope of length 2.4 m .



- (a) Find Simon's maximum speed as he swings.

(4)

- (b) Calculate the tension in the rope when Simon's speed is a maximum.

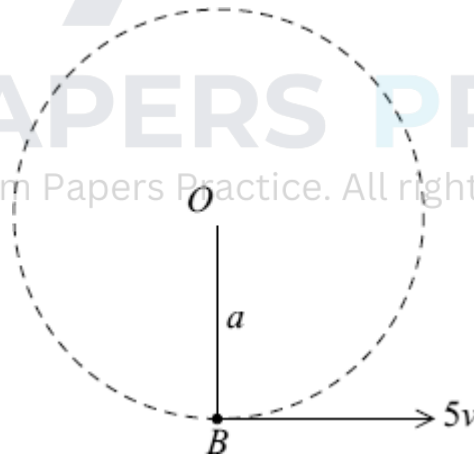
(3)

(Total 7 marks)

Q6.

A light inextensible string, of length a , has one end attached to a fixed point O .

A small bead, of mass m , is attached to the other end of the string. The bead is moving in a vertical circle, centre O . When the bead is at B , vertically below O , the string is taut and the bead is moving with speed $5v$.



- (a) The speed of the bead at the highest point of its path is $3v$.

Find v in terms of a and g .

(4)

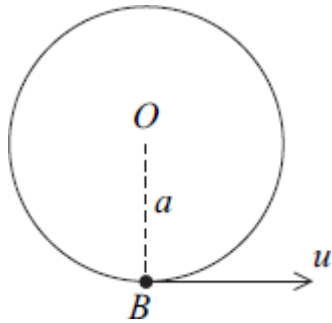
- (b) Find the ratio of the greatest tension to the least tension in the string, as the bead travels around its circular path.

(5)

(Total 9 marks)

Q7.

A smooth wire is fixed in a vertical plane so that it forms a circle of radius a metres and centre O . A bead, B , of mass 0.3 kg, is threaded on the wire and is set in motion with a speed u m s⁻¹ at the lowest point of its circular path, as shown in the diagram.

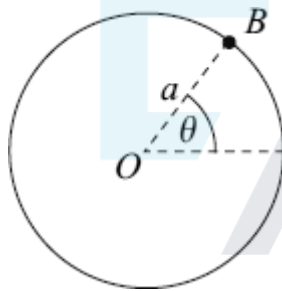


- (a) Show that, if the bead is going to make complete revolutions around the wire,

$$u > 2\sqrt{ag}$$

(3)

- (b) At time t seconds, the angle between OB and the horizontal is θ , as shown in the diagram.



It is given that $u = \sqrt{\frac{9}{2}ag}$.

- (i) Find the reaction of the bead on the wire, giving your answer in terms of g and θ .

(5)

- (ii) Find θ when this reaction is zero.

(2)

(Total 10 marks)

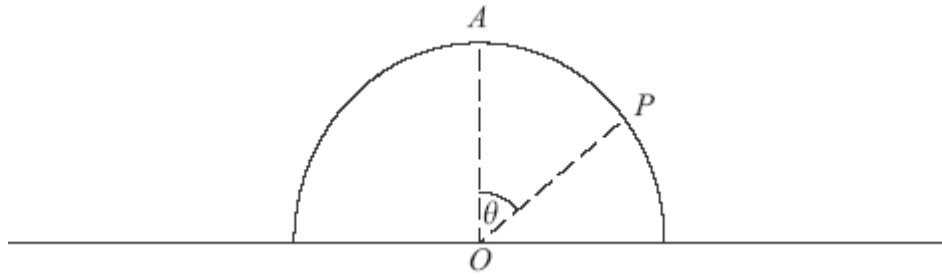
Q8.

A smooth hemisphere, of radius a and centre O , is fixed with its plane face on a horizontal surface. A particle, of mass m , can move freely on the surface of the hemisphere.

The particle is placed at the point A , the highest point of the hemisphere, and is set in motion along the surface with speed u .

- (a) While the particle is in contact with the hemisphere at a point P , OP makes an

angle θ with the upward vertical.



Show that the speed of the particle at P is

$$\left(u^2 + 2ga[1 - \cos \theta]\right)^{\frac{1}{2}}$$

(5)

- (b) The particle leaves the surface of the hemisphere when $\theta = \alpha$.

Find $\cos \alpha$ in terms of a , u and g .

(5)

(Total 10 marks)

Q9.

A particle is attached to one end of a light inextensible string of length 3 metres. The other end of the string is attached to a fixed point O . The particle is set into motion horizontally at point P with speed v , so that it describes part of a vertical circle whose centre is O . The point P is vertically below O .



The particle first comes momentarily to rest at the point Q , where OQ makes an angle of 15° to the vertical.

- (a) Find the value of v .

(4)

- (b) When the particle is at rest at the point Q , the tension in the string is 22 newtons.

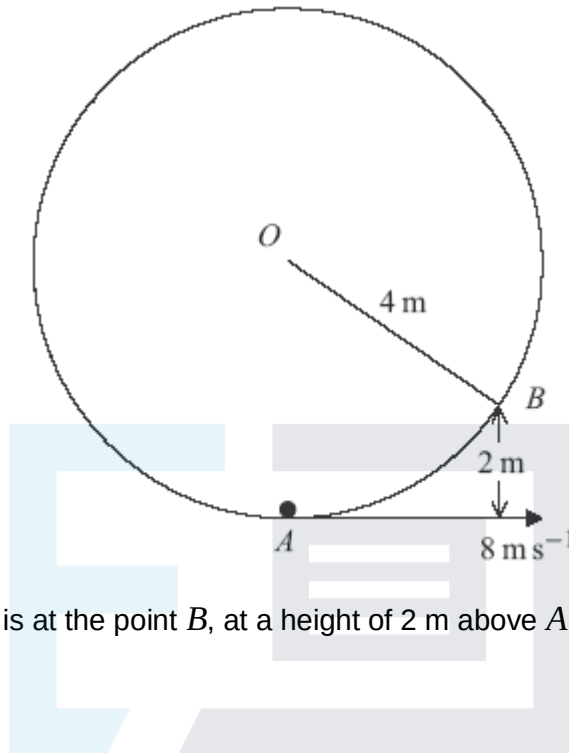
Find the mass of the particle.

(3)

(Total 7 marks)

Q10.

A hollow cylinder, of internal radius 4 m, is fixed so that its axis is horizontal. The point O is on this axis. A particle, of mass 6 kg, is set in motion so that it moves on the smooth **inner** surface of the cylinder in a vertical circle about O . Its speed at the point A , which is vertically below O , is 8 m s^{-1} .



When the particle is at the point B , at a height of 2 m above A , find:

- (a) its speed;
- (b) the normal reaction between the cylinder and the particle.

(3)

(4)

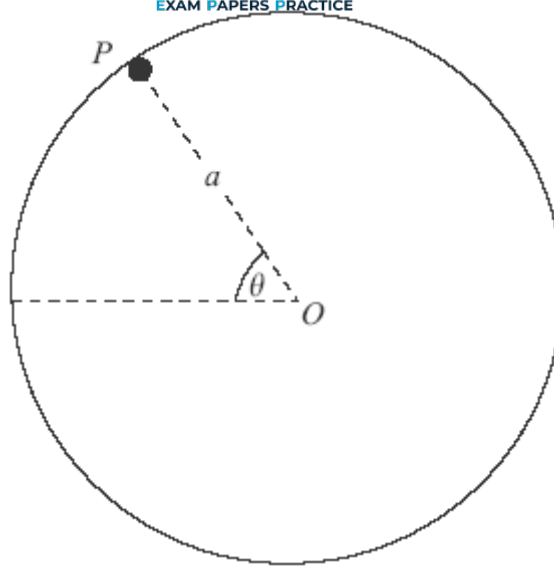
(Total 7 marks)

Q11.

In crazy golf, a golf ball is hit so that it starts to move in a vertical circle on the inside of a smooth cylinder.

Model the golf ball as a particle, P , of mass m . The circular path of the golf ball has radius a and centre O . At time t , the angle between OP and the horizontal is θ , as shown in the diagram.

The golf ball has speed u at the lowest point of its circular path.



- (a) Show that, while the golf ball is in contact with the cylinder, the reaction of the cylinder on the golf ball is

$$\frac{mu^2}{a} - 3mg \sin \theta - 2mg$$

(6)

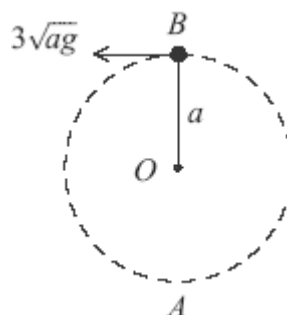
- (b) Given that $u = \sqrt{3ag}$, the golf ball will not complete a vertical circle inside the cylinder. Find the angle which OP makes with the horizontal when the golf ball leaves the surface of the cylinder.

(4)

(Total 10 marks)

Q12.

A light inextensible string, of length a , has one end attached to a fixed point O . A particle, of mass m , is attached to the other end. The particle is moving in a vertical circle, centre O . When the particle is at B , vertically above O , the string is taut and the particle is moving with speed $3\sqrt{ag}$.



- (a) Find, in terms of g and a , the speed of the particle at the lowest point, A , of its path.

(4)

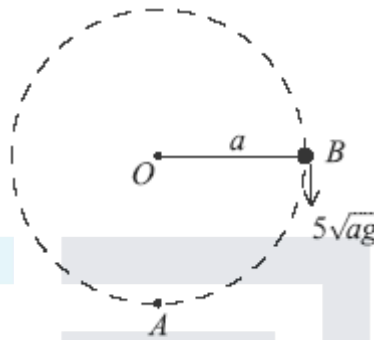
- (b) Find, in terms of g and m , the tension in the string when the particle is at A .

(4)

Q13.

A light inextensible string, of length a , has one end attached to a fixed point O . A particle, of mass m , is attached to the other end of the string. The particle is set into vertical circular motion with radius a and centre O .

When the particle is at B , on the same horizontal level as O , the string is taut and the particle is moving vertically downwards with speed $5\sqrt{ag}$.



- (a) Find, in terms of a and g , the speed of the particle at the lowest point, A , of its path.

(4)

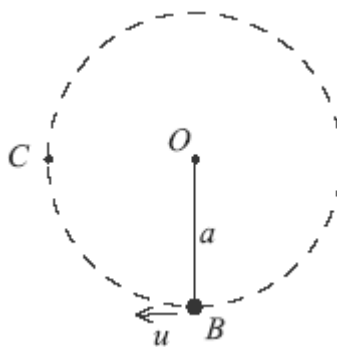
- (b) Find, in terms of m and g , the tension in the string when the particle is at A .

(3)

(Total 7 marks)

Q14.

A small bead, of mass m , is suspended from a fixed point O by a light inextensible string, of length a . The bead is then set into circular motion with the string taut at B , where B is vertically below O , with a horizontal speed u .



- (a) Given that the string does not become slack, show that the least value of u required for the bead to make complete revolutions about O is $\sqrt{5ag}$.

(5)

- (b) In the case where $u = \sqrt{5ag}$, find, in terms of g and m , the tension in the string when the bead is at the point C , which is at the same horizontal level as O , as

shown in the diagram.

(3)

- (c) State one modelling assumption that you have made in your solution.

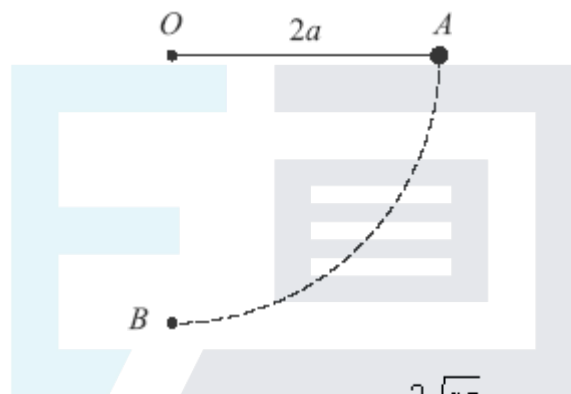
(1)

(Total 9 marks)

Q15.

A light inextensible string has length $2a$. One end of the string is attached to a fixed point O and a particle of mass m is attached to the other end. Initially, the particle is held at the point A with the string taut and horizontal. The particle is then released from rest and moves in a circular path. Subsequently, it passes through the point B , which is directly below O .

The points O , A and B are as shown in the diagram.



- (a) Show that the speed of the particle at B is $2\sqrt{ag}$.

(3)

- (b) Find the tension in the string as the particle passes through B . Give your answer in terms of m and g .

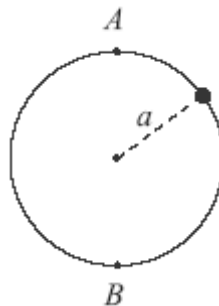
(3)

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(Total 6 marks)

Q16.

A bead of mass m moves on a smooth circular ring of radius a which is fixed in a vertical plane, as shown in the diagram. Its speed at A , the highest point of its path, is v and its speed at B , the lowest point of its path, is $7v$.



- (a) Show that $v = \sqrt{\frac{ag}{12}}$.

(5)

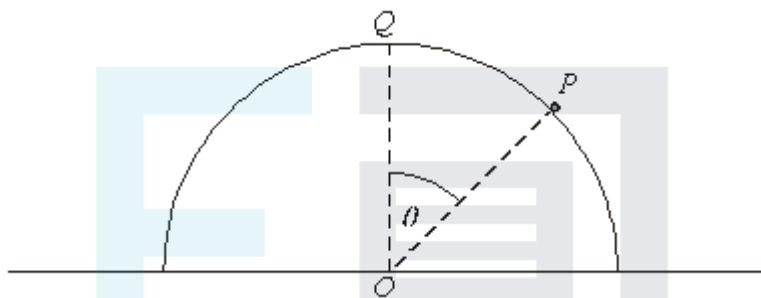
- (b) Find the reaction of the ring on the bead, in terms of m and g , when the bead is at A .

(4)

(Total 9 marks)

Q17.

A particle P , of mass m kg, is placed at the point Q on the top of a smooth upturned hemisphere of radius 3 metres and centre O . The plane face of the hemisphere is fixed to a horizontal table. The particle is set into motion with an initial horizontal velocity of 2 m s^{-1} . When the particle is on the surface of the hemisphere, the angle between OP and OQ is θ and the particle has speed $v \text{ m s}^{-1}$.



- (a) Show that $v^2 = 4 + 6g(1 - \cos \theta)$.

(4)

- (b) Find the value of θ when the particle leaves the hemisphere.

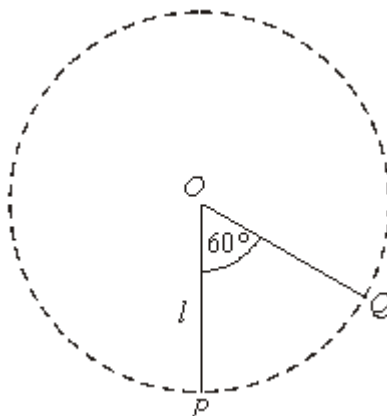
(5)

(Total 9 marks)

Q18.

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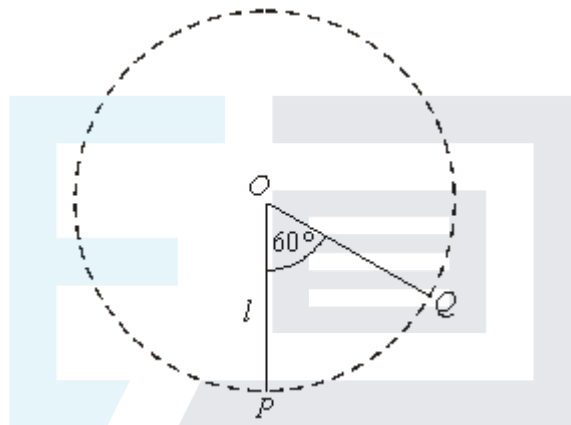
A particle of mass m is suspended from a fixed point O by a light inextensible string of length l . The particle hangs in equilibrium at the point P vertically below O . The particle is then set into motion with a horizontal velocity U so that it moves in a complete vertical circle with centre O . The point Q on the circle is such that $\angle POQ = 60^\circ$, as shown in the diagram.



- (a) Find, in terms of g , l and U , the speed of the particle at Q . (4)
- (b) Find, in terms of g , l , m and U , the tension in the string when the particle is at Q . (5)
- (Total 9 marks)

Q19.

A particle of mass m is suspended from a fixed point O by a light inextensible string of length l . The particle hangs in equilibrium at the point P vertically below O . The particle is then set into motion with a horizontal velocity U so that it moves in a complete vertical circle with centre O . The point Q on the circle is such that $\angle POQ = 60^\circ$, as shown in the diagram.

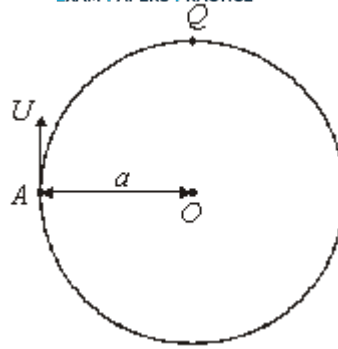


- (a) Find, in terms of g , l and U , the speed of the particle at Q . (4)
- (b) Find, in terms of g , l , m and U , the tension in the string when the particle is at Q . (5)
- (c) Find, in terms of g , l , m and U , the tension in the string when the particle returns to P . (2)

(Total 11 marks)

Q20.

A smooth circular wire, of centre O and radius a , is fixed in a vertical plane. A small bead, of mass m , is free to move on the wire. The bead is projected upwards, with initial speed U , from a point A on the wire which is level with O , as shown in the diagram.



- (a) Find, in terms of a and g , the least value of U required for the bead to reach the point Q , at the top of the circular wire, in the subsequent motion.

(3)

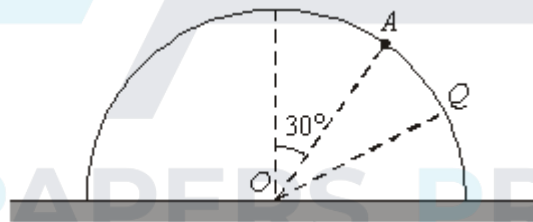
- (b) If $U = \sqrt{\frac{5ag}{2}}$, find the vertical height, h , above the level of O at which the magnitude of the normal reaction between the bead and the wire is zero.

(8)

(Total 11 marks)

Q21.

A solid smooth hemisphere, of radius r , has its plane face fixed to a horizontal table. The centre of the plane face is O .



A particle P , of mass m , is released from rest at a point A on the surface of the hemisphere, where OA makes an angle of 30° with the upward vertical through O . The particle P slides freely down the surface. At the point Q on the surface, the particle has speed v and OQ makes an angle θ with the upward vertical through O .

- (a) Show that $v^2 = gr(\sqrt{3} - 2\cos\theta)$.

(4)

- (b) Find, in terms of m , g and θ , an expression for the reaction force of the hemisphere on the particle at the point Q .

(5)

- (c) When $\theta = \alpha$, the particle loses contact with the hemisphere.

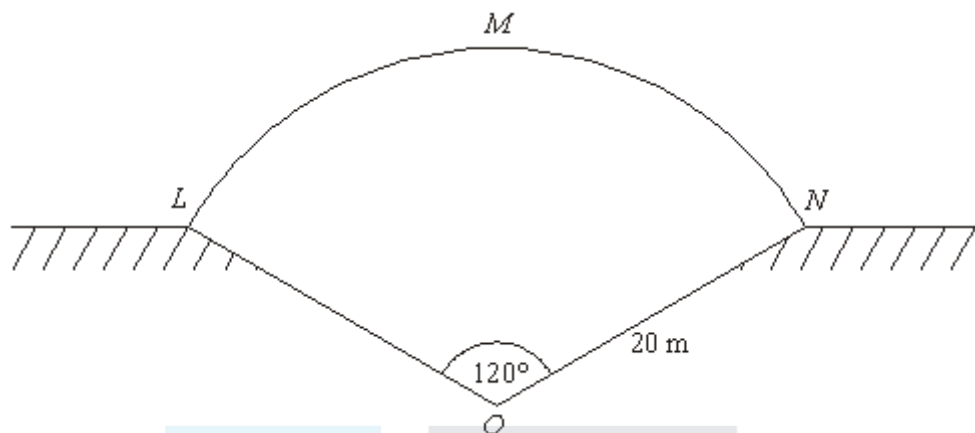
Find the value of α , giving your answer to the nearest degree.

(2)

(Total 11 marks)

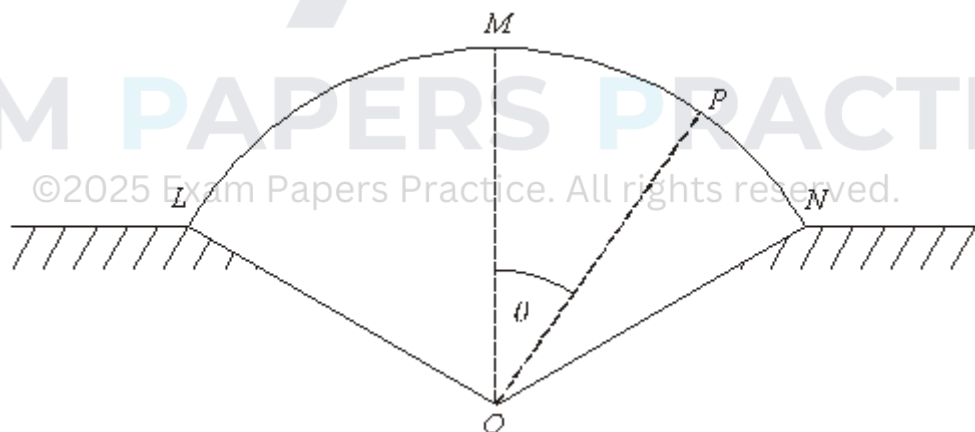
Q22.

A cyclist approaches a hump-backed bridge which is in the form of a circular arc LMN . The radius of the arc is 20 m and its centre is O . The angle between OL and ON is 120° . The points L and N are on the same horizontal level and M is the highest point of the bridge, as shown in the diagram.



The total mass of the cycle and cyclist is 60 kg. They may be modelled as a single particle. Assume that there is no resistance to the motion.

- (a) Calculate the gain in potential energy as the cyclist moves from L to M . (3)
- (b) At M the speed of the cyclist is 6 m s^{-1} . He begins to freewheel from M towards N . The point P is on the bridge between M and N and angle $MOP = \theta$, as shown in the diagram.

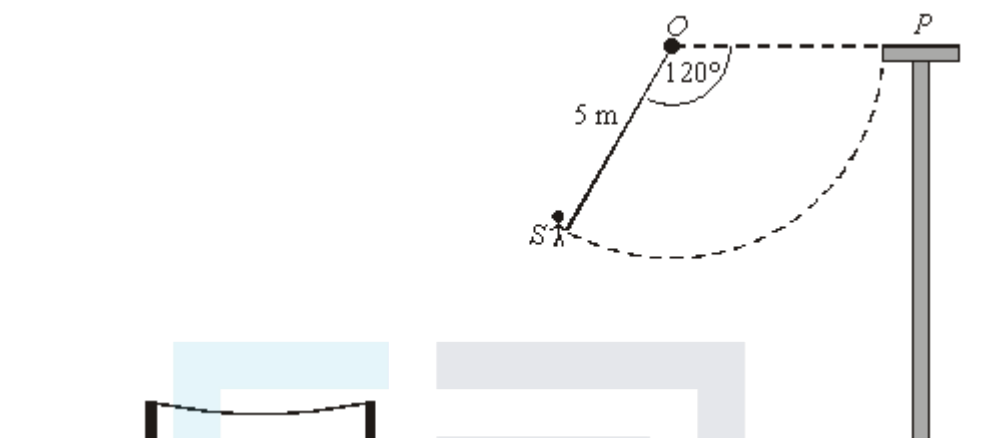


When the cyclist is at the point P his speed is v .

- (i) Show that $v^2 = 428 - 392 \cos \theta$. (4)
- (ii) Find, in terms of θ , the normal reaction between the bridge and the cycle at P . (4)
- (iii) Determine whether or not the cycle remains in contact with the bridge throughout the descent from M to N . (3)

Q23.

A stuntman S stands initially on a platform P . He holds a rope which is attached to a fixed support O . The points O and P are at the same horizontal level. The man then steps off the platform with the rope taut and horizontal. He swings through a circular arc, as shown in the diagram below.



The mass of the man is 70 kg and the length of the rope is 5 m . The man may be modelled as a particle throughout the motion.

- (a) The man releases his grip on the rope when it has swung through an angle of 120° .

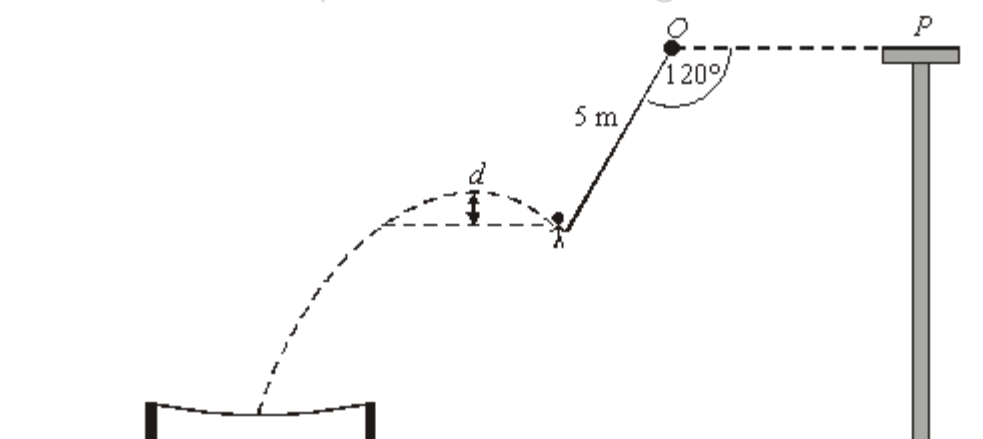
- (i) Show that the speed of the man when he is about to let go of the rope is approximately 9.21 m s^{-1} .

(4)

- (ii) Find the tension in the rope when the man is about to let go of the rope.

(5)

- (b) After letting go of the rope, the stuntman moves freely under gravity.



He rises a distance d before beginning to fall into the safety net. Show that d is approximately 1 metre .

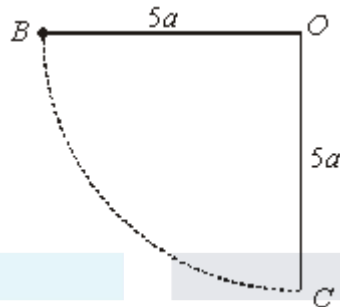
(4)

- (c) Comment on the assumption that the man can be modelled as a particle.

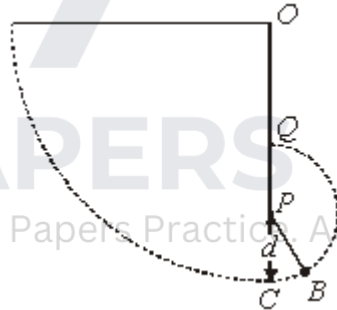
(1)
(Total 14 marks)

Q24.

Adam has set up an experiment for his Mechanics coursework. He has attached a small ball, B , of mass m , to one end of a light inextensible string of length $5a$. The other end of the string is attached to a fixed point O . The ball is released from rest with the string taut and horizontal, as shown in the diagram. The ball subsequently passes through the point C , which is a vertical distance $5a$ below O .



- (a) Find an expression, in terms of a and g , for the speed of B when it reaches C . (2)
- (b) A small smooth peg, P , is fixed at a distance d vertically above C . When the string reaches the vertical position, B begins to move in a vertical circle with centre P and radius d , as shown in the diagram.

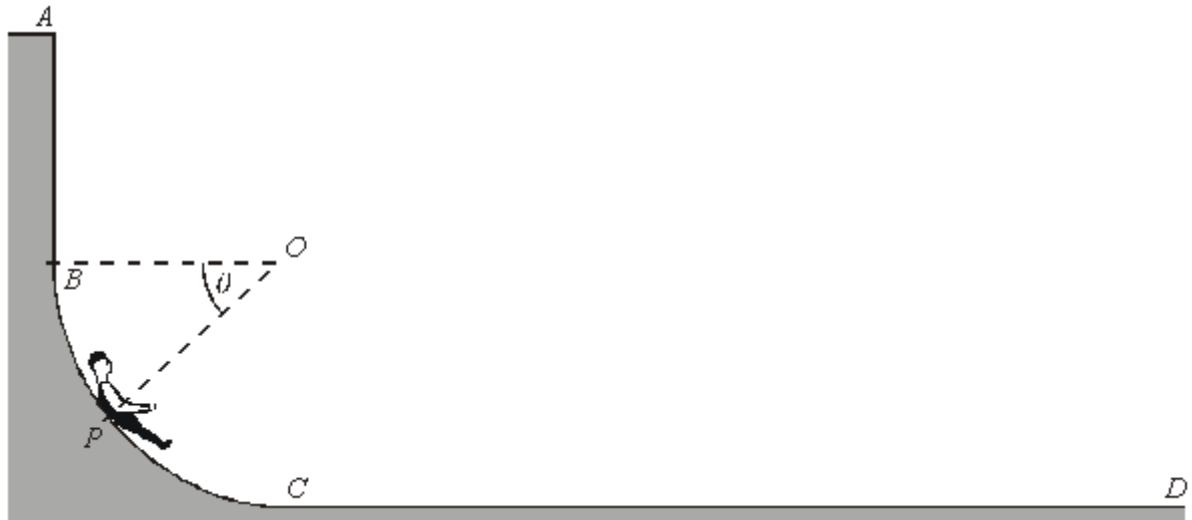


The ball reaches Q , the point at a distance d vertically above P , with speed v . At Q , the string is taut.

- (i) Show that $v^2 = 2g(5a - 2d)$. (4)
- (ii) Find, in terms of a , d , g and m , the tension in the string when the ball is at Q . (4)
- (iii) Hence show that $d < 2a$. (2)
- (c) State **one** modelling assumption used in this question. (1)

(Total 13 marks)

Q25.



The diagram shows a vertical cross section of a new adventure slide at a theme park. It consists of three sections AB , BC and CD .

Section AB is smooth and vertical and has length r .

Section BC is smooth and forms a quarter of a circle. This circle has centre O and radius r . The radius OB is horizontal and OC is vertical.

Section CD is rough, straight and horizontal. It is of length $4r$.

Steve, who has mass m , starts from rest at A and reaches speed u at the point B . He remains in contact with the surface until he reaches D .

It can be assumed that Steve can be modelled as a particle throughout the motion.

- (a) Find u^2 in terms of g and r . (2)
- (b) Steve reaches the point P between B and C where angle $POB = \theta$, as shown in the diagram. His speed at P is v .
 - (i) Show that $v^2 = 2gr(1 + \sin \theta)$. (4)
 - (ii) Draw a diagram showing the forces acting on Steve when he is at the point P . (1)
 - (iii) Find an expression for the normal reaction, R , on Steve when he is at the point P . Give your answer in terms of m , g and θ . (4)
- (c) Show that, as Steve crosses C , there is a reduction in the normal reaction of magnitude $4mg$. (2)

- (d) Between C and D , Steve decelerates uniformly and comes to rest at the point D .

Find his retardation.

(3)

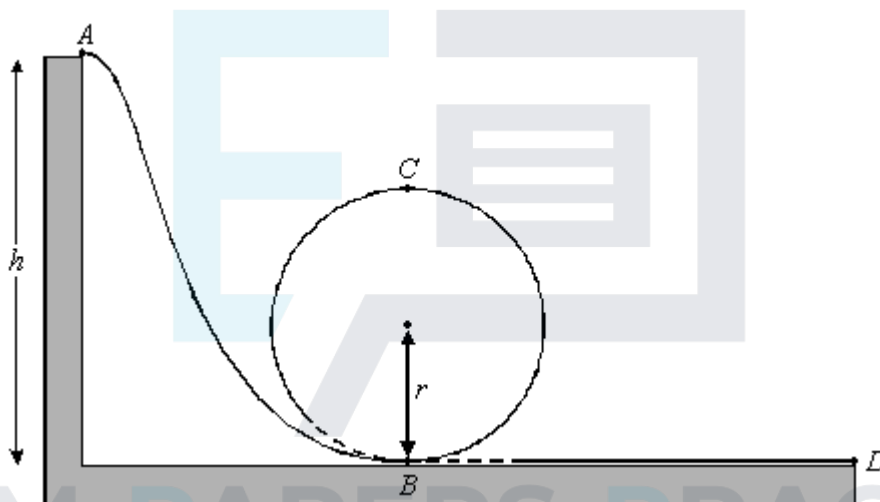
(Total 16 marks)

Q26.

Maria is modelling the motion of a toy car along a “loop the loop” track.

The track, $ABCD$, can be modelled as a continuous smooth surface contained in a vertical plane. The highest point on the track is A , which is a distance h above the floor. The loop is a circle of radius r with BC as a diameter, and with C vertically above B . The track ends at the point D , as shown in the diagram.

Maria is trying to determine a connection between h and r in the case where the car stays only just in contact with the track at C .



Maria models the car as a particle of mass m , which starts from rest at A . In the case where the car stays only just in contact with the track at C , it has speed u at B and speed v at C .

- (a) By considering the forces on the car at C , show that $v^2 = rg$.

(3)

- (b) Find an expression for u^2 in terms of g and r .

(4)

- (c) Show that $h = kr$, where k is a constant to be determined.

(3)

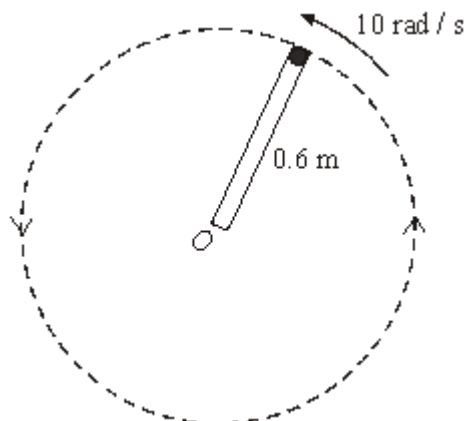
- (d) Suggest one improvement that could be made to the model in order to refine the solution.

(1)

(Total 11 marks)

Q27.

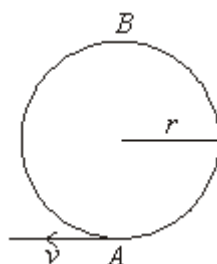
The diagram shows a smooth tube, sealed at each end, which rotates about one end. There is a small sphere inside the tube. The tube rotates in a vertical plane, so that the sphere remains at the end of the tube and describes a vertical circle of radius 0.6 metres. The mass of the sphere is 0.05 kg and the tube rotates with a constant angular speed of 10 radians per second.



- (a) Calculate the magnitude of the acceleration of the sphere. (2)
 - (b) Find the magnitude of the horizontal reaction force exerted by the end of the tube on the sphere when the tube is horizontal. (2)
 - (c) Find the magnitude of the vertical reaction force exerted by the end of the tube on the sphere when the sphere is at its lowest point. (3)
- (Total 7 marks)

Q28.

A bead, of mass m , is threaded onto a smooth circular ring, of radius r , which is fixed in a vertical plane. The bead is moving on the wire. Its speed, v , at the lowest point of its path, A , is five times its speed at the highest point, B .



- (a) Show that $v = 5\sqrt{\frac{gr}{6}}$. (4)
- (b) Find the reaction of the wire on the bead, in terms of m and g , when the bead is:
 - (i) at B ; (3)

- (ii) at a height of $\frac{1}{2}r$ above A .

(5)

(Total 12 marks)

Q29.

A smooth hemisphere of radius a and centre Q lies with its plane face fixed to a horizontal surface. A particle of mass m can move freely on the surface of the hemisphere.

Initially, the particle is at rest at the highest point of the hemisphere. It is set in motion along the surface of the hemisphere. The line QP makes an angle of θ with the upward vertical.



- (a) Show that, when the particle is at P , its speed is

$$\sqrt{2ga(1 - \cos \theta)}$$

(3)

- (b) Find the value of $\cos \theta$ if the particle leaves the surface of the hemisphere when it reaches P .

(4)

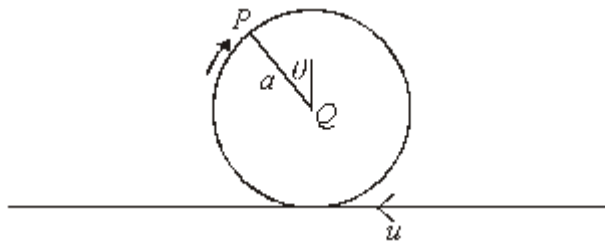
(Total 7 marks)

Q30.

In crazy golf, a golf ball is fired along a smooth track and loops the loop inside a section of track.

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Model this loop as a vertical circle of radius a and centre Q , as shown in the diagram.



The golf ball is travelling at speed u as it enters the circle at the lowest point.

Model the ball as a particle P , of mass m .

- (a) Show that the reaction of the track on the particle when QP makes an angle of θ with the upward vertical is

$$\frac{mu^2}{a} - 3mg \cos \theta - 2mg$$

(6)

- (b) Given that the ball completes a vertical circle inside the track, show that

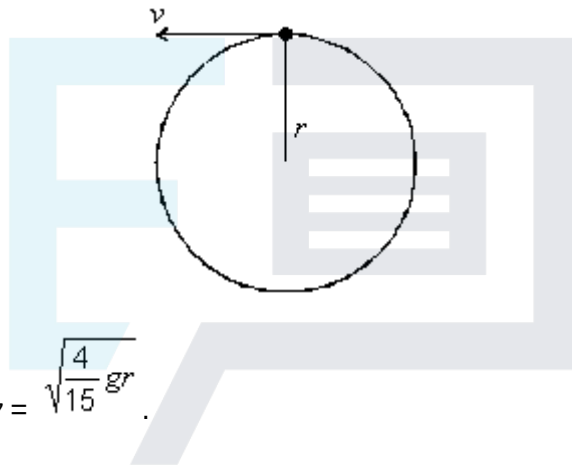
$$u \geq \sqrt{5ag}$$

(2)

(Total 8 marks)

Q31.

A bead, of mass m , is threaded onto a smooth circular ring, of radius r , which is fixed in a vertical plane. The bead is moving on the wire. Its speed, v , at the highest point of its path is one quarter of its speed at the lowest point.



- (a) Show that $v = \sqrt{\frac{4}{15}gr}$.

(4)

- (b) Find the reaction of the wire on the bead, in terms of m and g , when the bead is:

- (i) at the highest point;

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(3)

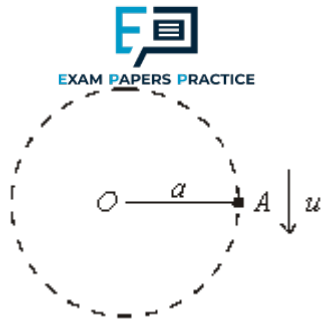
- (ii) $\frac{1}{2}r$ below its highest point.

(5)

(Total 12 marks)

Q32.

One end of a light inextensible string of length a is attached to a fixed point, O , and a particle of mass m is attached to the other end, A . The particle is held so that the string is taut and OA is horizontal. It is then projected vertically downwards with speed u as shown in the diagram.



The string becomes slack when OA is inclined at an angle of 60° above the horizontal.

- (a) Show that the speed of the particle when the string becomes slack is $\sqrt{\frac{\sqrt{3}}{2}ag}$.

(3)

- (b) Hence find u in terms of a and g .

(4)

(Total 7 marks)



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